

IMPULSE: Important Message Prioritization Using Relevancy Scoring Engine

Kunal Pasad¹, Riti Vakil¹, Khushi Soni¹, Pramod Bide¹

¹Computer Engineering, Sardar Patel Institute of Technology, Munshi
Nagar, Mumbai, 400058, Maharashtra, India.

Contributing authors: kunal.pasad@spit.ac.in; riti.vakil@spit.ac.in;
khushi.soni@spit.ac.in; pramod.bide@spit.ac.in;

Abstract

In today's digital age, our reliance on digital messaging platforms, especially apps like WhatsApp, has grown exponentially. These platforms are not just conduits for personal communication but have also become crucial for disseminating news and updates. However, this convenience comes with challenge: the overwhelming influx of messages can lead to critical information being missed, especially when users return to a flood of unread messages after a period of inactivity. To address this issue, we recognize the pressing need for advance technology that intelligently classifies and highlights messages based on their importance. By developing IMPULSE (Important Message Prioritization Using Relevancy Scoring Engine), a model for text classification, we aim to ensure that important messages are notified, highlighted and displayed to the user. This innovative approach will enhance the user experience, allowing individuals to prioritize and engage with significant messages amidst the constant flow of digital communication.

Keywords: Message overload, message prioritization, relevancy classification, push notifications, highlighting mechanism.

1 Introduction

Examining the regularity with which mobile devices emit notifications to signify the reception of new messages reveals a widespread trend. Numerous individuals find their smartphones persistently abuzz with an incessant inflow of text messages, WhatsApp communications, and assorted notifications, thereby establishing a continuous stream of informational stimuli[1].

Recent investigations conducted by the Pew Research Center provide compelling statistical insights[2]. In 2015, their findings indicated that over one-third of smartphone users, constituting approximately 36%, availed themselves of messaging applications such as WhatsApp. Furthermore, in the year 2011, their research unveiled that adults engaged in an average of 42 daily text exchanges. Notably, within the demographic of individuals aged 18 to 29 years, the frequency surged significantly, with an average of approximately 88 daily text interactions.

Also according to a survey[3] on smartphone usage in the 21st century: who is active on WhatsApp The mean of typical daily smartphone usage was 161.95 min (SD = 83.36). The mean of typical daily WhatsApp usage was 32.11 min (SD = 35.36). Thus, WhatsApp alone accounted for about 20% (19.83%) of typical daily smartphone usage in the present sample. These numerical representations underscore the pervasive integration of messaging into contemporary lifestyles. However, the proliferation of messages poses a notable challenge in terms of managing and staying abreast of communication. The substantial volume of messages occasionally leads to the inadvertent oversight of critical information. This issue holds significant ramifications, particularly concerning the potential omission of essential messages from familial connections or crucial work-related updates.

This study addresses the need for efficient management of WhatsApp messages through the development of an intelligent system. Our research endeavors to devise a methodology that scrutinizes unread messages, discerning their significance based on textual content. Conceptually, this system functions akin to an adept assistant, systematically prioritizing and presenting important messages at the forefront using context aware notifications[4].

The implementation involves the use of classification algorithms[5] to achieve our results. These models analyze the linguistic content of messages to make informed decisions regarding their immediate relevance. Such an approach holds promise as a valuable resource for individuals contending with message overload, offering a potential remedy for those grappling with the daily influx of messages.

In this chapter, we will discuss how we developed this system, how it operates, and why it offers a promising solution to the issue of handling numerous messages. We anticipate that our solution presents a viable and efficacious approach to mitigate the challenges associated with the overwhelming volume of WhatsApp messages, ultimately simplifying and alleviating the user experience.

2 Literature Review

In today’s digital landscape, messaging platforms like WhatsApp have become integral to our daily communication, facilitating instant exchanges of information. However, amidst the deluge of messages and interruptions[6], distinguishing between what’s important and what’s not can be a daunting task. This challenge has spurred a growing interest in text classification, a field dedicated to automating the categorization of messages based on their relevance to users.

Gelbukh’s seminal survey[7] from 2011 laid the groundwork for understanding text classification algorithms, exploring both traditional methodologies and modern

machine learning approaches. This foundational work provided a springboard for subsequent research, serving as a compass for those navigating the complexities of message prioritization within digital communication channels.

Building upon Gelbukh’s framework, a more recent study[5] by Kamran Kowsari and colleagues in 2019 offered a deeper dive into text classification techniques. Their comprehensive survey traversed various domains, from sentiment analysis to spam detection, shedding light on the strengths and limitations of each approach. This survey became an invaluable resource for researchers and practitioners alike, guiding the selection of appropriate methods for specific classification tasks.

A groundbreaking approach, Top2Vec[8] by Dima Angelov, addresses longstanding challenges faced by traditional techniques. By leveraging document embeddings and hierarchical clustering, Top2Vec offers dynamic topic identification without the constraints of predefined topic numbers. Its scalability and efficiency in handling large text datasets, coupled with superior performance in tasks like document clustering and topic summarization, mark a significant advancement in the field of natural language processing.

Furthermore, efforts have been directed towards implementing intelligent notification systems that prioritize information based on user relevance. This research[9] introduces the Awareness and Notification Service (ANS), operating on a rule-based approach where users specify conditions for notifications using a rule language. ANS components include EventMonitor, Controller, and Notifier, facilitating context data gathering, rule evaluation, and tailored notification delivery. ANS rule language supports complex event relations and enables developers to define notification triggers based on context changes. This streamlines the development of context-aware applications by automating context data management and notification processes.

A deeper research[10], by Swadhin Pradhan and fellow researchers, investigates notification usage patterns and proposes a notification management system. With users receiving an average of 60 notifications per day, mainly during waking hours, and spending about 15 minutes daily checking notifications, there is a clear need for effective management. The proposed system predicts notification importance with 87% accuracy using contextual information and past user interactions, aiming to prioritize notifications based on importance to potentially improve user experience. Such systems are crucial in alleviating notification fatigue and ensuring that users are only alerted to what matters most, ultimately enhancing the overall communication experience.

Despite the advancements in text classification, there exists a paucity of research specifically tailored to the unique characteristics of WhatsApp messages. Informal language, frequent use of emojis, and varying message lengths pose challenges that necessitate a more context-aware approach.

Furthermore, existing literature often overlooks the real-time nature of WhatsApp conversations, highlighting the need for models capable of adapting quickly to evolving conversational dynamics. Moreover, the exploration of multimodal approaches that consider both textual and visual elements remains relatively unexplored, presenting an opportunity for future research to bridge this gap in knowledge.

3 Proposed Methodology

In our pursuit to notify users only when messages of specific relevance arrive, we have undertaken a methodical approach grounded in data-driven methodologies and ethical considerations. Leveraging Python3 alongside NLTK libraries[11], we have crafted a comprehensive methodology that encompasses data collection, preprocessing, feature extraction, model selection, and ethical guidelines.

3.1 Data Extraction

Initially, we extracted a substantial corpus from our personal WhatsApp conversations. This strategic selection was predicated on the premise that incorporating real-world, user-generated content would significantly enrich the dataset with genuine conversational contexts. We integrated additional messages obtained from publicly accessible online repositories. This deliberate amalgamation was designed to ensure a balanced representation of conversation dynamics. The dataset was meticulously curated to comprise approximately 9,400 messages, maintaining an equitable distribution between our original chat data and the externally sourced messages, each constituting 50% of the total corpus.

3.2 Data Preprocessing and Transformation

We embarked on a rigorous data preprocessing phase to ensure the dataset’s readiness for analytical processing. The key steps for preprocessing[12] were Column Renaming, Handling Missing Values, Removing Duplicate Entries, Data Integrity Verification, Target Variable Distribution. The text transformation and cleaning stage was meticulously designed to refine the textual data, enhancing its suitability for machine learning analysis. The key steps[13] were Tokenization, Removal of Non-Alphanumeric Characters, Exclusion of Stopwords and Punctuation, Stemming. A function encapsulates these steps, systematically converting raw text into a format more amenable to machine learning analysis. By meticulously cleaning and transforming the text, we significantly improve the data quality, ensuring that our model is trained on relevant and standardized textual input. This process is pivotal for enhancing model accuracy, as it allows the algorithm to learn from a distilled representation of the text.

3.3 Top2Vec

We delineated a novel classification criterion tailored to the unique requirements of our chat message dataset. This criterion was predicated on the thematic categorization of messages into distinct sectors such as health, education, law, IT, food, etc. To operationalize this classification, we embarked on a meticulous process of lexicon construction using Top2Vec[8] for extraction of keywords based on topics. Top2Vec is a sophisticated algorithm designed to streamline topic modeling and semantic search processes with remarkable efficiency. At its core, Top2Vec excels in automatically detecting underlying topics within a body of text, enabling users to uncover meaningful insights without the need for extensive manual intervention. Its versatility lies in its ability to generate jointly embedded vectors for topics, documents, and words,

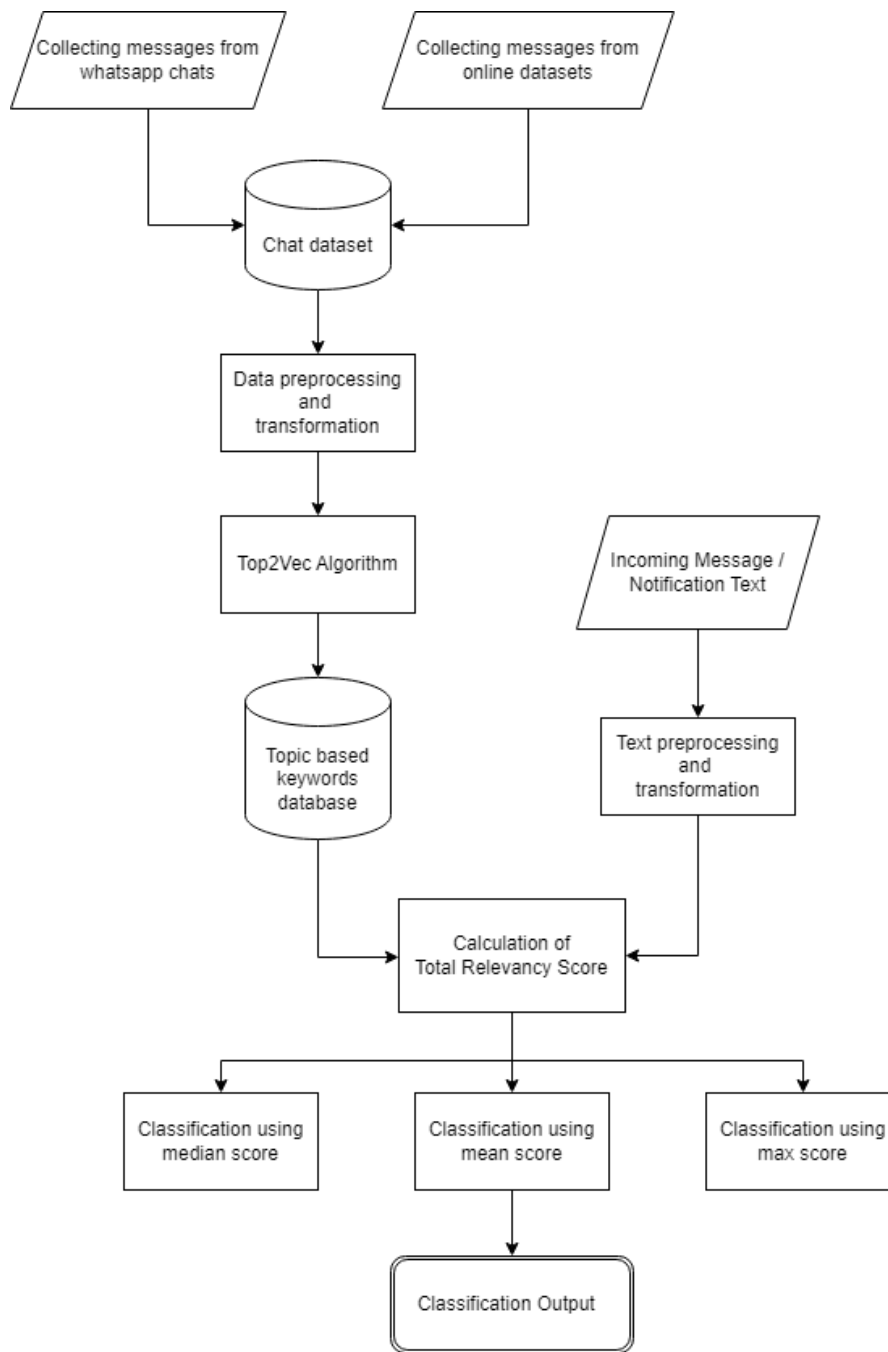


Fig. 1 Architecture Of IMPULSE

effectively capturing the intricate relationships within textual data. Top2Vec can be considered as an algorithm for performing topic modeling in a very easy way. Top2Vec is an algorithm for topic modeling and Semantic search. It automatically detects the topics present in the text and generates jointly embedded topic, document and word vectors. After training the Top2Vec model we get a number of detected topics, can search topics by keywords and find similar words. It includes the generation of embedding vectors that allows us to represent the text document in the mathematical framework. This framework can be multi-dimensional where the dimension depends on the word or text document. The generated high dimensional vectors get reduced in their size. This is a basic dimension reduction process and Top2Vec uses the UMAP dimension reduction technique[14]. This allows us to find a dense area for clustering. Clustering is facilitated through the DBSCAN technique[15], which partitions the dimensionally reduced vectors into distinct groups, each representing a cluster of related topics or documents. We calculate the centroid of every dense area of the clusters. Then it finds the n-closest word vectors and feeds them to the topic vector so that they can become topic words.

3.4 Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer

Upon the completion of the phase, we proceeded to the pivotal stage of feature extraction. This step is instrumental in translating the processed text into a numerical format that machine learning models can interpret and analyze effectively. A key technique employed in this process was the utilization of the Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer[16], a sophisticated method that quantifies text while capturing the importance of words within the dataset. The TF-IDF Vectorizer transforms the cleaned text into a matrix of TF-IDF features. This method calculates the frequency of each word in a message relative to its frequency across all messages in the corpus, thereby highlighting words that are unique to certain messages. Such a weighting scheme enhances the model’s ability to discern and prioritize words that contribute significant informational value over common terms that appear frequently across texts. The choice of parameters for the TF-IDF Vectorizer, particularly `max_features`, is critical in optimizing the feature extraction process. The `max_features` parameter restricts the number of most informative words to be considered, effectively setting an upper limit on the vocabulary size. This strategic selection is driven by the dual goals of reducing computational complexity and focusing the model’s attention on the most impactful features. By limiting the feature set to the top terms as defined by `max_features`, we ensure that the model is trained on words that offer the greatest relevance for analysis, thereby improving efficiency without compromising the quality of insights derived.[16]

3.5 Relevancy Classification

We quantitatively assessed words on a scale of 0 to 1 based on their frequency in sector-specific discourse, capturing their relative importance. For each message, we

computed the mean of these values to gauge thematic alignment and importance. Messages with a mean score above 0.7 were labeled 'important,' signifying strong sectoral resonance. This binary classification system formed our dataset labeling, enabling systematic categorization based on sectoral relevance. Rigorous exploration of alternative statistical measures and threshold values led us to adopt the mean value with a 0.7 threshold, which showed the highest accuracy in distinguishing between 'important' and 'not important' messages.

3.6 Model Training and Selection

We began our model selection process by evaluating three Naive Bayes classifiers[17], aiming to leverage their probabilistic foundations for text classification. Among them, the Bernoulli Naive Bayes (BNB) emerged as the most effective due to its suitability for handling binary feature representations common in text data. BNB set a baseline for performance benchmarks in our study.

Expanding our analysis, we explored a range of machine learning algorithms[7] to identify models that could surpass BNB's performance. This comparative analysis revealed the XGBoost (XGB) model[18] as the optimal choice, boasting an impressive accuracy of 96.55% and a precision of 98.25%. Other models such as the AdaBoost Classifier (ABC), Logistic Regression (LRC), and Support Vector Classifier (SVC) also demonstrated substantial efficacy, showcasing the competitive nature of our analysis[19].

Conversely, models like the Bagging Classifier (BC), Random Forest Classifier (RFC), and Extra Trees Classifier (ETC) and [20]Nearest Neighbors (KN) delivered moderate results and were outperformed in our specific context. This underscores the importance of tailored model selection[19]. The XGBoost model's effectiveness can be attributed to its sophisticated optimization techniques, including regularization mechanisms that mitigate overfitting, and its ability to handle various data structures and distributions efficiently.

In conclusion, our research journey encompassed comprehensive data collection, rigorous preprocessing, advanced text transformation, model selection, and a novel classification criterion. Ethical considerations were integral, reflecting our commitment to responsible research practices. The resulting model and classification scheme not only optimize analytical performance but also contribute to the broader discourse on ethical challenges in text-based communication analysis.

3.7 Notifications Based Messaging

We integrated notification delivery[6] strategies within the context of WhatsApp, aiming to identify important messages from unread chats. The approach involves the adaptation of the in-app and push notification framework[21] to WhatsApp's messaging interface. The ANS service primarily keeps the applications aware of context and environmental changes, while applications register monitoring rules specifying which changes in context should trigger notifications to users. The environmental changes are modeled using event condition action (ECA) rules. Unread chats serve as the focal point for identifying potentially significant messages. Firstly, the system evaluates

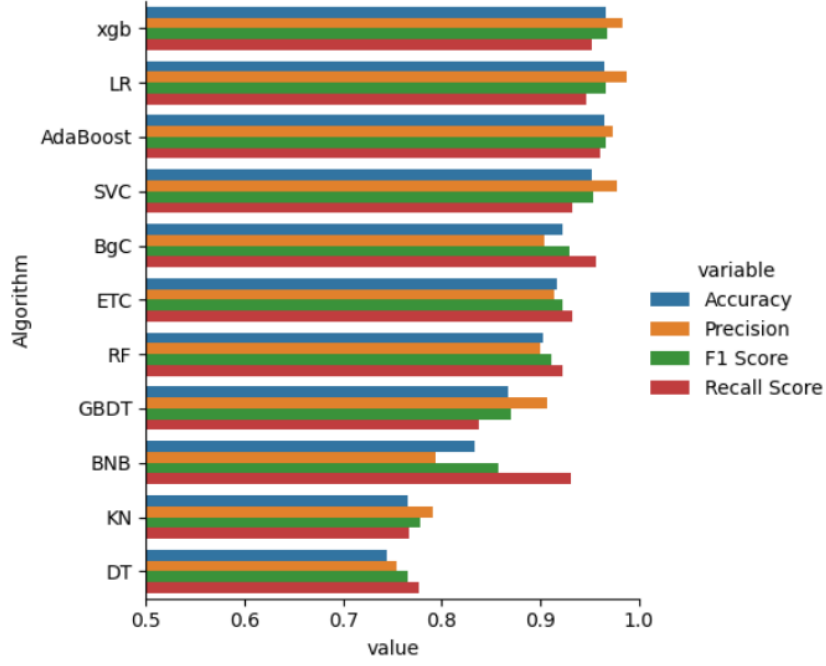


Fig. 2 Graph Comparison between Models

whether a message qualifies as important based on criteria. If a message is deemed important, the message is flagged for further action. The decision-making process for message prioritization encompasses two sequential steps: determining if the message warrants in-app notification within the WhatsApp interface and subsequently evaluating whether it merits a push notification to the user's device. These decisions are informed by considerations of message relevance, urgency, and user preferences. The methodology, termed as channel selection within the WhatsApp environment, guides the prioritization and delivery of notifications to enhance user engagement and streamline communication effectiveness. Once a flagged message is received, the system implements a highlighting mechanism within the WhatsApp interface. Upon opening the chat containing the important message, the message is visually highlighted to draw the user's attention. This visual cue serves as a prompt for the user to engage with the message promptly. Simultaneously, if the important message remains unread for a specified period, typically indicated by a predefined time threshold, a push notification[21] is triggered to alert the user of the unread important message.

4 Results and Discussion

The Relevancy Score for messages ranged between 0 and 1 with an average selected of 0.70 by us as the threshold. A sample sample from the classified data along with its corresponding label and total relevancy score. The model was tested across 9400 randomly selected text messages from the Gitter dataset. Out of these, 421 messages

were classified incorrectly. The developed system successfully addressed the challenge of managing overwhelming WhatsApp message volumes by intelligently classifying and highlighting important messages. Leveraging advanced text classification techniques, particularly the Top2Vec algorithm[8] for thematic categorization and the XGBoost model[18] for message prioritization, the system achieved an impressive accuracy of 96.55% and a precision of 98.25%. By systematically categorizing messages based on sectoral relevance, the system ensured that users were promptly notified and presented with significant messages amidst the continuous flow of digital communication. In our evaluation, Table 1 summarizes the performance metrics, including Accuracy, Precision, F1 Score, and Recall, of various models applied to the classification task.

Algorithm	Accuracy	Precision	F1 Score	Recall
xgb	0.965517	0.982548	0.967354	0.952623
LR	0.964610	0.987633	0.966292	0.945854
AdaBoost	0.964610	0.972603	0.966809	0.961083
SVC	0.951906	0.976950	0.954113	0.932318
BgC	0.921960	0.904000	0.929276	0.956007
ETC	0.916515	0.913765	0.922948	0.932318
RF	0.902904	0.899340	0.910610	0.922166
GBDT	0.866606	0.906593	0.870712	0.837563
NB	0.833031	0.793651	0.856698	0.930626
KN	0.765880	0.790576	0.778351	0.766497
DT	0.744102	0.753695	0.765000	0.776650

Table 1 Accuracy, Precision, F1 Score and Recall of Different Models

5 Conclusion

In summary, this research presents an innovative solution to tackle the daunting task of managing the ever-increasing volume of WhatsApp messages. Through the application of intelligent classification and prioritization techniques, as demonstrated in Table 1, we showcase the superior performance of various algorithms. By harnessing advanced text classification methods and introducing novel approaches such as Top2Vec and XGBoost, our system adeptly discerns and emphasizes crucial messages within the continuous stream of digital communication. Throughout our methodology, we have prioritized ethical considerations, ensuring responsible research practices are integral to our approach. This commitment underscores our dedication to conducting research that is both effective and conscientious. In essence, our system represents a significant advancement in the domain of messaging platforms, providing a practical and efficient remedy to the challenges posed by message overload. By enhancing the user experience and simplifying message management, our approach facilitates more streamlined and meaningful digital interactions.

Furthermore, it's worth noting that our research not only addresses the immediate obstacles presented by message overload but also lays the foundation for future innovations in digital communication management. As technology continues to reshape our modes of interaction, our approach stands ready to revolutionize how we engage with messaging platforms, promoting efficiency, clarity, and ultimately enriching the digital experience for users worldwide.

6 Future Scope

Looking ahead, the potential for advancements in this domain is immense. Future research avenues include exploring real-time classification mechanisms, integrating textual and visual cues through multimodal approaches, and implementing personalized prioritization based on user feedback. Extending the application of our system to diverse messaging platforms and addressing unique challenges could significantly amplify its impact. While our current model supports only English text, expanding it to encompass multiple languages[22] and integrating image processing for document and image classification would enhance its capabilities. Additionally, the development of a semantic-based, context-aware notification system[23] could further optimize user experience by tailoring notifications to individual preferences, fostering more efficient and intuitive communication experiences.

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